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StructAgent: Orchestrating Cryo-EM Model Building and Refinement with a Multi-Agent LLM System

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Abstract

Building and refining cryo-EM atomic models often requires long, project-specific workflows that combine map inspection, prior structural knowledge, restraints, refinement, validation and expert review. Existing programs perform many individual operations, but coordinating them across iterative model-building sessions remains manual and difficult to audit. We present StructAgent, a user-guided multi-agent resource for cryo-EM model building and refinement. StructAgent couples a domain agent for literature-grounded structural reasoning with an execution agent that runs local software, tracks state, recovers from failures and records provenance. Expert approval gates control major model-changing actions. In three case studies, StructAgent refitted a 64-chain proteasome from an earlier template, audited 530 ribosomal metal-ion sites and guided a chemically ambiguous ligand fit in a folate-metabolism enzyme from ongoing work. These demonstrations show that agentic orchestration can convert modeling intent into auditable, reviewable software workflows while preserving expert control and final scientific judgment.

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Cryo-electron microscopy (cryo-EM) has become central to the structural analysis of large, flexible and heterogeneous macromolecular assemblies. In many current projects, atomic model building no longer starts from a single map and an empty canvas. Structural biologists work with global and focused maps, predicted and homologous starting models, ligand or ion hypotheses, biochemical constraints and prior structural literature. Compared with typical crystallographic maps at comparable interpretive stages, cryo-EM maps are often noisier, have lower or more variable local resolution, and provide less direct evidence for side chains, ligands and ions. Turning these inputs into a reliable model therefore requires adaptive sequences of fitting, rebuilding, restraint generation, refinement, validation and expert review. Mature programs support many individual steps, including automated model building [1-6], interactive flexible fitting [7] and reciprocal- or real-space refinement [5,8,9].

The remaining bottleneck is workflow coordination. Cryo-EM model building is rarely a linear run of one program: the user must choose tools, set parameters, inspect intermediate outputs, recover from failures, compare alternative strategies and document why each step was taken. This burden is especially large for multi-chain assemblies, flexible conformational states, focused refinements, difficult ligand chemistry and large-scale local validation tasks. The unmet need is therefore not only to generate coordinates, but to make complex model-building workflows easier to explore, audit and repeat. A structural biologist may want to test whether domain-wise fitting improves a flexible assembly, whether a different refinement route better preserves geometry, whether a ligand orientation is chemically defensible, or whether hundreds of metal-ion sites should be re-evaluated under explicit coordination assumptions.

Large language models (LLMs) provide a possible interface for this type of workflow orchestration because they can translate natural-language intent into plans, commands and tool

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calls [10-18]. Related agent systems have begun to coordinate scientific tools in chemistry, drug discovery and protein-science settings [19-23]. General-purpose LLM use, however, is not sufficient for structural biology. Atomic model building is a high-consequence scientific task: hallucinated commands, unsupported interpretations, undocumented parameters, lost context or unreviewed model changes can compromise reproducibility. Long cryo-EM sessions add a further challenge because tool logs, file paths, errors and validation outputs can accumulate over hours or days. An LLM-based system for cryo-EM therefore needs a constrained operating model, not only conversational fluency.

We developed StructAgent as a user-guided orchestration resource for cryo-EM model building and refinement. StructAgent separates scientific reasoning from execution management through two cooperating agents. Maria, the domain agent, evaluates strategies using a curated structural-biology knowledge base, literature retrieval and project memory. Annika, the execution agent, translates approved strategies into local software actions, manages files and state, monitors outputs, diagnoses failures and records provenance. Both agents operate through editable skill documents that encode tool use, workflow logic, validation checks and recovery procedures for programs such as ChimeraX, ISOLDE, PHENIX, Coot, Servalcat/REFMAC5 and gemmi. The trust model is deliberately conventional: established software performs the calculations; skills and literature constrain reasoning; major model-changing actions require expert approval; and runs leave logs, metrics, inputs, outputs and rationale for later review.

The operational unit of the system is the skill. Skills are readable protocol documents rather than hidden prompts: they specify expected inputs, command patterns, validation metrics, common failure modes and recovery actions. This makes the resource inspectable and editable by the same researchers who rely on it. The system can therefore accumulate workflow knowledge

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without turning it into an opaque automation layer; a user can review how a tool was called, why a recovery action was chosen and which parts of the protocol should be revised after a failed or successful run.

We evaluated StructAgent using three representative cryo-EM task classes. In a 64-chain proteasome example, StructAgent coordinated domain-aware refitting, restraint generation, flexible fitting, refinement and validation starting from an earlier structural template. In a 1.55 Å ribosome example, it built and executed a systematic audit of 530 metal-ion sites and refined 75 user-approved corrections with tailored restraints. In a ligand-fitting example, it combined local density fitting with literature-grounded reasoning to choose the orientation of a folate-derived ligand in an ambiguous binding site. These examples are case studies, not formal benchmarks or universal protocols. They test whether an agentic system can coordinate multi-tool cryo-EM workflows, recover from execution failures and provide evidence packages for expert review. StructAgent is not a coordinate-prediction algorithm or an autonomous substitute for structural biology expertise; it is an auditable layer that helps experts convert modeling intent into reviewable computational workflows.

Results

System overview

StructAgent is a conversational workflow system for cryo-EM model building and refinement (Fig. 1). The user communicates in natural language through standard messaging applications, while all structural-biology software runs locally. Maria provides literature-grounded domain reasoning and validation criteria. Annika manages local execution, retry logic, workflow state, validation output and provenance capture. The two-agent design is intended to keep execution-

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heavy context separate from scientific reasoning during long sessions; inter-agent consultations are logged through the A2A channel and described in Online Methods. StructAgent uses editable skill protocols (Supplementary Table 1), a curated knowledge base (Supplementary Table 2), established structural-biology software, approval gates and session logs as its audit framework. The output is not an autonomous final structure but a reviewable package of models, maps, metrics, logs and rationale from which the structural biologist chooses the next action.

Representative cryo-EM task classes

The three examples below stress different aspects of StructAgent: large-assembly rebuilding, systematic local validation and chemically informed ligand fitting. They demonstrate rapid, auditable exploration of model-building ideas rather than benchmark superiority over expert manual workflows.

Example 1: whole-assembly rebuilding of the 26S proteasome

We first tested whether StructAgent could coordinate a whole-assembly rebuild rather than a single local fitting decision. The target was the human 26S proteasome map EMD-8332 at 3.8 Å, with 64 chains, 32 unique subunits and 19,957 residues in the rebuilt model. As the starting scaffold, we used PDB 5GJR, an earlier related 26S proteasome model that provided chain topology but was a poor rigid-body fit to this map. Deposited PDB 5T0C, the expert-refined model for EMD-8332, was held out from the workflow and used only as a post hoc reference.

Fig. 2a summarizes the division of work. The user supplied the input files and rebuilding goal; Maria analyzed the inputs, wrote the staged plan and later reviewed the refinement results; Annika executed the approved workflow. The task required domain definition, chain anchoring, restraint generation, flexible fitting, reciprocal-space refinement, real-space refinement and

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validation across all 64 chains. Annika generated the chain-specific fitting scripts using Merizo [24] for domain segmentation and ProSMART [25] restraints derived from AlphaFold [26] models, then ran the workflow with ChimeraX, ISOLDE, Servalcat/REFMAC5, PHENIX, gemmi and supporting scripts.

The workflow did not succeed by running a fixed script end to end. An initial AlphaFold-only assembly route produced unstable inter-domain geometry, so Maria re-planned the rebuild around 5GJR as a topology scaffold and AlphaFold monomers as domain and restraint references. Annika then rigid-fit 5GJR into the map, refit each chain through its best-matching domain anchor, assembled the full complex and carried the model through ISOLDE flexible fitting, Servalcat/REFMAC5 refinement and PHENIX real-space refinement. During this process Annika recovered five tracked execution failures, including force-field parameterization gaps, a hydrogenation-induced coordinate-frame shift, post-simulation chain drift, a case-insensitive chain-file collision on macOS and B-factor inflation after refinement (Supplementary Table 3).

The reported StructAgent rebuild converted a weak template fit into a strong expert-handover model (Fig. 2b). The figure compares three states: the rigid-body 5GJR template fit, the StructAgent rebuild and the deposited 5T0C reference, all validated against EMD-8332. CC_mask increased from 0.413 to 0.705, EMRinger from 0.091 to 0.662, rotamer outliers decreased from 11.54% to 1.78% and clashscore from 63.67 to 17.31. Under the validation pipeline used here, the rebuild also exceeded deposited 5T0C on density-agreement metrics, including CC_mask (0.705 versus 0.594), EMRinger (0.662 versus 0.562) and FSC = 0.5 (4.80 versus 6.45 Å). This is workflow evidence, not a claim that the rebuilt model supersedes 5T0C: validation used the deposited full map because half-maps were unavailable, and stronger density-fit scores may reflect overfitting risk or differences in modern restraints.

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The comparison also defined the boundary of the automation. Local geometry improved substantially from the 5GJR template, but deposited 5T0C remained better on rotamer outliers (0.16% versus 1.78%), Ramachandran outliers (0.85% versus 2.19%), C® deviations and bond/angle RMSD. The rebuild's better Rama-Z whole score (-0.78 versus -4.16 for 5T0C) suggests that some validation differences reflect newer refinement libraries, but it does not remove the need for expert inspection of local side-chain choices at 3.8 Å.

The per-chain analysis in Fig. 2c shows that the improvement was broad rather than driven by one selected region. All 64 chains improved in map agreement relative to the rigid-body 5GJR fit, with mean ⊗CC = +0.24 and median ⊗CC = +0.23. The local examples in Fig. 2d make the same point visually: Rpn5 improved from CC 0.29 to 0.58, and Rpt2 from 0.31 to 0.75. In the post hoc comparison with deposited 5T0C, the rebuild had higher per-chain CC for 61 of 64 chains, with only small regressions in the remaining three.

The endpoint was therefore expert handover, not autonomous deposition. StructAgent had completed the coordination-intensive part of the rebuild, captured the failures and recovery steps, and produced a reviewable model package. The remaining work was the part that should remain with a structural biologist: targeted side-chain correction, local geometry review and final scientific judgment.

Example 2: systematic metal-ion audit of a 1.55 Å ribosome

Metal-ion identities in cryo-EM models are often difficult to assign because Coulomb potential maps do not provide the anomalous signal available in X-ray crystallography. An experienced crystallographer can inspect a few suspected sites by comparing coordination distances, ligand types and expected geometry with tabulated values [27], but this process does not scale cleanly

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to large assemblies. PDB 8B0X, a 1.55 Å *E. coli* 70S ribosome [28], contains 530 deposited metal sites across 104 chains. Majorek et al. [29] had already reported four misassigned ions in this structure using the CheckMyMetal (CMM) validation server. We therefore asked whether StructAgent could turn classical coordination chemistry into a systematic, auditable audit, correction and restraint-generation workflow for every site, rather than a manual review of selected examples.

The first constraint came from the user rather than from the coordinates alone: Mg^{2+}, K^{+} and Zn^{2+} were specified as candidate identities based on the buffer composition and known ribosomal chemistry. This experimental context limited the search to chemically plausible ions before any scoring was done. Maria then translated bond-valence-sum (BVS) parameter sets [30,31] together with Harding's [27] metal-ligand coordination geometry into an executable specification, including R_0 parameters, expected coordination numbers, distance cutoffs, ligand-type checks and confidence grades. Annika implemented the audit with gemmi by extracting each coordination shell, scoring every site against the candidate ions and assigning grades from A to F. The full 530-site audit ran in a single session and separated plausible assignments from sites requiring review (Fig. 3a). After user review, 75 corrections were approved: 56 Mg^{2+} to K^{+}, 10 Mg^{2+} to Zn^{2+}, 2 Mg^{2+} to water and 7 K^{+} to water.

The audit did not stop at a list of suspicious ions. Annika generated ion-specific coordination restraints for the corrected model and refined it with REFMAC5 through Servalcat. This step moved the workflow beyond what a validation server alone provides: the same literature-derived assumptions used to flag sites were converted into restraint files for refinement across more than 500 local environments. The most difficult recovery involved external angle restraints. Servalcat's chain-renaming scheme produced nonpolymer identifiers that REFMAC5 could not

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parse through the `EXTE ANGL` route. After five failed formulations, Maria recast the angle restraints as equivalent 1-3 distance restraints using the law of cosines, and Annika executed the refinement through `EXTE DIST`. This restraint cascade accounts for five of the eight Example 2 recovery events in Supplementary Table 3.

Independent CMM [32] rescoring supported the corrected model while keeping the comparison anchored to an established validation tool (Fig. 3b,c). CMM was applied to the original 530 deposited positions; after correction, the 9 positions reclassified as water were excluded from corrected ion-score totals, leaving 521 ion sites. Top-scoring CMM sites increased from 69% to 79%, score-0 outliers decreased from 15 to 4, R-factor improved from 0.3836 to 0.3443 and FSC_work improved from 0.7864 to 0.8342. The local examples in Fig. 3d show both kinds of outcome: retained Mg^{2+} sites with improved coordination geometry and a Mg^{2+} to K^{+} reassignment at A/3044, one of the sites previously documented by Majorek et al. [29].

This case is not presented as a universal metal-curation protocol or as a critique of the original deposition. The Zn^{2+} reassignments remain the least independently supported class, consistent with atypical or incomplete peripheral coordination environments. The demonstration is instead that StructAgent could combine user-supplied experimental context, literature-grounded BVS rules, large-scale coordinate analysis, restraint generation, refinement and independent validation into one reviewable workflow. Conventional tools scored sites or refined coordinates; the agentic contribution was connecting those operations into a chemically reasoned strategy at a scale that is impractical to perform by hand.

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Example 3: ligand fitting with literature-grounded orientation reasoning

Ligand fitting in cryo-EM often requires more than maximizing map correlation. At typical local resolutions, ligand density can be incomplete, flexible groups may be weak or disordered, and multiple chemically different poses can give plausible local fits. We therefore tested whether StructAgent could use prior structural knowledge to guide ligand placement while still keeping the decision auditable. The target was a bifunctional folate-metabolism enzyme from an ongoing cryo-EM study. The unliganded 3.2 Å focused map lacked 5-formimidoyltetrahydrofolate (CHEBI:57456), an approximately 18 Å folate derivative, at two sites. One site, the transferase active site, had a direct 1.7 Å X-ray homolog (PDB 1QD1, ligand FON). The second site, at the inter-domain interface, had no homologous ligand pose and showed elongated density compatible with two orientations.

The executable workflow is summarized in Fig. 4a. Annika routed the task through existing tools for restraint generation, homolog transfer, flexible fitting, real-space refinement and validation, and recovered an ISOLDE/OpenMM template failure by generating a USER_LIG force-field file (Supplementary Table 3). The figure carries the tool sequence; the central result is the decision logic for the inter-domain interface site.

Maria evaluated the ambiguous site by combining residue chemistry, folate-binding literature and substrate-channeling logic, rather than by asking which orientation gave the highest density score alone. She recommended the pterin-to-chain-C orientation because Tyr419 could stack the pterin ring, Arg424, Arg436 and Asp423 could support pterin recognition, and the Lys193-His197-Arg198-Arg381 positive cluster faced the glutamate carboxylates. ChimeraX `fitmap` supported this choice but did not by itself settle it: the selected orientation had z_mean 7.35

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versus 6.06 for the alternative (⊗z_mean = 1.29). At Maria's request, Annika then ran a per-group comparison on unrefined seed poses; the selected orientation was better in every ligand group, including pterin-ring CC 0.19 versus 0.04 and formimidoyl z-score 9.27 versus 4.68.

Maria also shaped the validation endpoint. The first combined PHENIX refinement placed both ligands, but diagnostics showed frozen ligand B-factors and pterin-ring planarity distortion. Maria advised a single combined PHENIX pass with individual isotropic ADP refinement and custom pterin planarity restraints, while explicitly rejecting manual B-factor rescaling and over-restraining of the flexible glutamate tail. This raised CC_mask from 0.7997 to 0.8194, improved the inter-domain interface ligand CC from 0.5848 to 0.6785 and gave the transferase active-site ligand CC of 0.6713 (Fig. 4b,c).

The final evidence package reported both support and uncertainty. HOLE identified a continuous channel between the FT and CD sites with a minimum radius of 6.09 Å, consistent with the selected orientation. The Site 1 glutamate tail remained weak, with local CC approximately 0.09, consistent with a surface-exposed flexible tail at 3.2 Å; this was reported rather than forced into density. This case therefore highlights the role of Maria: conventional software generated poses, scores and refined coordinates, but the decisive contribution was literature-grounded chemical interpretation, diagnostic review and restraint choice.

Cross-case summary

Across the three case studies, StructAgent coordinated workflows using ChimeraX, ISOLDE, PHENIX/eLBOW, Servalcat/REFMAC5, gemmi, Coot and custom validation scripts; recovered all 14 tracked tool-level failures; and produced reviewable evidence packages containing models, maps, metrics, logs and rationale. The agent architectures supporting this division of domain

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reasoning and execution management are summarized in Fig. 5. Fig. 6 illustrates why this separation matters during long workflows, where execution history can otherwise compete with domain-level reasoning. The corrected failure distribution is 5/5 for the proteasome rebuild, 8/8 for the metal-ion audit and 1/1 for ligand fitting (Supplementary Table 3). Most recoveries were handled by Annika without user intervention; the hardest cases required Maria's domain reasoning or explicit user approval for model-changing decisions. The case studies also illustrate how successful recovery patterns can be distilled into future skill updates rather than remaining as one-off troubleshooting episodes.

Discussion

StructAgent addresses a practical bottleneck in cryo-EM model building: not the absence of capable fitting, refinement or validation programs, but the difficulty of coordinating them across long, project-specific workflows. The system is therefore best understood as an auditable orchestration resource. It helps a structural biologist turn a modeling question into software actions, intermediate checks, recovery steps and evidence packages, while keeping model-changing decisions and final interpretation under expert control.

The three examples make complementary claims about this orchestration layer. The proteasome rebuild tests scale and expert handover: StructAgent coordinated a 64-chain rebuild from an earlier related template, improved all chains relative to the rigid-body template fit and produced a model suitable for targeted expert review. The ribosome example tests scalable strategy construction: Maria translated bond-valence and coordination chemistry into an audit specification, and Annika applied it to 530 metal sites, generated restraints and refined 75 user-approved corrections. The ongoing-work ligand example tests scientific reasoning: conventional

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tools could generate poses and density scores, but Maria connected residue chemistry, homologous folate binding and substrate-channeling logic to a defensible orientation choice, then defined a validation endpoint that preserved weak-tail uncertainty. Together, these examples show that the useful unit is not a single tool call, but a reviewable chain of reasoning, execution and validation.

StructAgent's added value is the connection between established tools. ChimeraX, ISOLDE, PHENIX, Servalcat/REFMAC5, Coot, gemmi and validation servers perform the structural calculations. StructAgent links them into workflows, records how they were used, monitors intermediate outcomes, recognizes when downstream symptoms indicate broken assumptions and preserves repairs as reusable workflow knowledge. Across the case studies, all 14 tracked tool-level failures were recovered, but the count is less important than the form of the recovery: coordinate-frame shifts, force-field template gaps, restraint-format incompatibilities and refinement artifacts became inspectable events with causes, fixes and provenance, rather than hidden troubleshooting steps in a manual session.

This provenance addresses a broader documentation problem in structural biology. Published methods sections often name the software used for model building and refinement, but rarely capture the actual command sequence, parameter changes, failed branches, diagnostic checks or rationale that shaped the final model. StructAgent produces session records containing tool invocations, inputs, outputs, validation metrics, errors and recovery decisions, with software versions captured where available. These records do not make LLM-guided orchestration deterministic, and they do not remove the risk of a plausible but wrong recommendation. They make the workflow inspectable: a reviewer or future user can ask what was run, why a route changed, which alternatives failed and what evidence supported a model-changing decision.

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Trust in StructAgent comes from constraint rather than autonomy. Atomic calculations are delegated to established structural-biology programs. Literature and skill documents constrain the reasoning space. Strategy changes, ligand-orientation choices, ion reassignments, residue mutations, broad refinement-route changes and deposition-related actions require user approval. These safeguards are not equivalent to formal verification, and they cannot guarantee that every recommendation is scientifically correct. They do, however, make the system closer to a disciplined and documented operator than to a black-box structure generator.

Skills are the reusable unit of this resource. A skill is not just a prompt fragment; it is a readable protocol that specifies expected inputs, command patterns, validation checks, common failure modes and recovery actions. Skills allow operational expertise to be revised after real use and reused in future sessions. The Maria-Annika separation is one implementation of this idea, designed to keep execution-heavy context, logs and file-state details separate from strategy, literature grounding and validation criteria. It is not a fixed requirement of the resource. The same skill documents could be used by a single sufficiently capable agent, by a different multi-agent arrangement, or by a general coding assistant that can read protocols and operate local files. Additional specialist agents, for example focused on biological interpretation, could be added without changing the principle that skills and logs are the portable substrate.

More generally, an agent does not know all experimental context: sample preparation, construct design, biochemical expectations, map-processing choices, unpublished controls and project priorities may be available only to the researcher. StructAgent can organize evidence and propose actions, but it should not be trusted blindly to decide what is biologically plausible or deposition-ready. The next step is therefore not to remove the expert from model building, but to reduce repetitive protocol execution. As more skills are generated, tested and revised, routine

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refinement setup, validation checks, simple fitting trials and repeated audit procedures should become increasingly agent-executable with limited intervention, while unusual chemistry, ambiguous density, weak local resolution and biologically consequential model changes continue to require explicit expert review. Future evaluation should combine public archives with installable code, sanitized skills, versioned command wrappers, smoke-test data and releasable case-study bundles; external installation by independent users; and prospective comparisons with manual or semi-automated workflows. StructAgent should not be presented as an autonomous deposition engine or a replacement for structural biology expertise. Its contribution is narrower and more practical: it preserves expert agency while making complex cryo-EM model-building workflows easier to execute, inspect, repeat and improve.

Online Methods

Overall system design

StructAgent is a domain-specific agent system for cryo-EM model building and refinement, built on a substantially modified version of the OpenClaw multi-agent framework (https://github.com/openclaw/openclaw). The base framework provides session management, tool execution and workspace state. StructAgent extends it with custom agent roles, memory handling, domain-specific inter-agent communication, structural-biology tool integration and session management for long, information-dense workflows.

The language model is embedded in an operational environment with explicit tools, bounded task protocols, persistent state and structured communication channels. The system is therefore used as a workflow controller rather than as a free-form chatbot. All structural-biology calculations are delegated to local software.

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Agent roles

Maria is the domain-specialist agent. She interprets scientific questions, evaluates strategies, grounds recommendations in literature and project memory, diagnoses domain-level failure modes and defines validation criteria. Her resources include a three-tier structural-biology knowledge base, literature-retrieval tools and project-specific memory.

Annika is the execution-specialist agent. She translates user intent or Maria-approved strategies into concrete software actions, sequences tool invocations, manages workflow state and handles error recovery. Annika's tool layer covers ChimeraX, ISOLDE, PHENIX, Coot, Servalcat/REFMAC5, gemmi and supporting scripts. When a task spans multiple programs, Annika composes the relevant skills dynamically and records the run state.

The internal architectures of Maria and Annika are summarized in Fig. 5.

Knowledge base

Maria's curated knowledge base is organized in three tiers, designed for efficient retrieval within token-budget constraints.

Tier 1 — project overview (approximately 500 tokens, always loaded). A lightweight brief captures the current scientific focus, key open questions and recent findings. This is loaded at the start of every session, giving Maria immediate project context.

Tier 2 — topic-specific knowledge files (approximately 1-3k tokens each, loaded on demand). Distilled summaries are organized by topic: method comparison tables, resolution-dependent parameter guidelines, decision trees for common tasks and operational lessons. These encode synthesized, actionable rules, for example, "B-factor refinement should be deferred until

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geometry is stable" or "maps below 4 Å typically require tighter secondary-structure restraints", rather than individual paper summaries. Each rule traces back to its source papers.

Tier 3 — full paper archive (searched by script, never bulk-loaded). Individual paper digests, structured metadata and a cross-reference index capture relationships between papers. The archive currently contains more than 80 papers and is queried through targeted searches. A full inventory is provided in Supplementary Table 2.

This three-tier design keeps Maria's token cost approximately flat as the knowledge base grows: adding more papers enriches Tier 3 and occasionally distills new rules into Tier 2, while the always-loaded context remains small.

When a question falls outside the curated database, Maria can extend retrieval through a literature-discovery skill that queries the Semantic Scholar API [33], deduplicates results against the existing database and triages candidates for relevance. New papers can be digested during a session or queued for later reading. The goal is to ground recommendations in curated or retrievable evidence rather than unsupported model memory.

Skills

Skills are human-readable protocol documents that define how a class of tasks should be performed: required inputs, tool commands, parameter choices, validation checks, expected outputs and common failure-recovery procedures. Because skills are readable and versioned, they convert tacit operational expertise into a reusable and auditable resource.

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Skill architecture

StructAgent uses a dedicated skill-creation pipeline with a top-level classifier that identifies the type of skill needed and applies the appropriate architecture. **Tool-interface skills** (for example, ChimeraX, PHENIX and ISOLDE) encode software-specific knowledge: command syntax, parameter conventions, input/output formats and known quirks. **Workflow-composition skills** (for example, structural-build and structural-strategy) encode higher-level decision logic: which tools to use for which problems, in what order and with what validation gates. **Knowledge-acquisition skills** (for example, paper-reader and discovery) encode processes for building and maintaining the knowledge base itself.

How skills are built

Tool-interface skills were assembled from official documentation, methodology papers, tutorials, observed command behavior and task logs, then revised after use on real projects. The resulting skill document is tested against real tasks, and failures drive iterative revision. Skills evolve through reviewed operational experience rather than through autonomous self-modification.

Skill evolution through use

The ISOLDE skill provides a concrete example. The initial version could launch ISOLDE sessions but had no robust way to monitor progress, detect convergence or catch failures; GUI state changes were invisible to the agent. Through iterative debugging across real tasks, the skill acquired (i) a Python monitoring bridge that reads simulation state from a running ISOLDE session at fixed intervals; (ii) explicit input-file verification, checking atom naming, chain assignments and residue completeness before launch, after discovering that ISOLDE fails silently on certain input errors; (iii) a pre-launch model-preparation checklist; and (iv)

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convergence criteria calibrated to different resolution ranges. At the end of sessions, the system can summarize completed work and propose reusable lessons; deeper lesson-distillation steps extract candidate updates to skills or knowledge files, such as new preflight checks, failure signatures or parameter rules. These updates are stored as proposed changes and presented for user review before being applied; the system does not autonomously modify its own protocols.

A full skill inventory for both agents is provided in Supplementary Table 1.

Execution tracking and inter-agent communication

During multi-step tasks, Annika records tool invocations, software versions where available, input files, command-line parameters, standard output and error streams, output files and validation metrics, including MolProbity-based geometry checks [34]. These logs allow interrupted workflows to be resumed, prevent repeated failed approaches and provide provenance records for completed runs.

A worked example is ISOLDE convergence monitoring. ISOLDE is an interactive molecular-dynamics flexible-fitting tool designed for human use in a 3D graphical environment. Making it agent-controllable required bridging the gap between the LLM's text-based reasoning and a GUI tool that assumes visual feedback. StructAgent uses Python monitoring scripts that read simulation state from the running ISOLDE session at fixed intervals, extract quantitative metrics (CC_mask, geometry statistics, simulation energy), detect stalls or failures and report structured status back to the agent. The agent can therefore launch an ISOLDE session, monitor its progress through numbers rather than visual inspection and terminate refinement when the change between consecutive CC_mask readings falls below a convergence threshold. The same pattern,

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a polling bridge that converts GUI state into structured status, is reused by other interactive tools in Annika's stack.

Long model-building sessions can accumulate enough operational detail to interfere with scientific reasoning. StructAgent addresses this with context compaction adapted from Lossless Context Management [35], by keeping execution history in Annika's context and by sending focused scientific questions to Maria only when domain reasoning is needed (Fig. 6). Inter-agent communication is implemented through the Agent-to-Agent (A2A) protocol [36], adapted with sender-side logging, long-running consultation timeouts and structured messages that separate scientific questions from operational metadata. Because Maria and Annika can run on different LLM backends, and Maria in particular can draw on more than one model, inter-agent consultation also functions as a form of cross-model reasoning, bringing complementary strengths to bear on the same scientific question.

Memory architecture

StructAgent uses a multi-timescale memory architecture: active session memory, periodic context compaction, end-of-session reflection and lesson distillation.

Active session memory

During a session, every action, tool output, decision and inter-agent consultation is recorded in a structured session log. This provides the operational history for the current task, supports backtracking and forms the raw material for downstream memory processes.

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Periodic context compaction

Long cryo-EM modeling sessions can run for hours, accumulating more context than the LLM's active processing limit can usefully hold. StructAgent uses periodic context compaction, an LLM-based process that summarizes completed portions of the session while preserving key decisions, metrics and outcomes; the original, uncompacted history remains stored and can be retrieved when needed. This mechanism builds on the Lossless Context Management (LCM) system [35], with custom compaction strategies tuned for the information density of structural-biology sessions, where a single tool output may contain hundreds of lines of validation statistics, only a few of which matter for future decisions.

End-of-session reflection (session distill)

At the end of each working session, the system runs a structured reflection step: it reviews what was accomplished, what problems were encountered, what decisions were made and what remains to be done, and writes a compact working-state file that the next session loads on startup. This is structured synthesis rather than automatic compaction, closer to a researcher writing up lab notes at the end of the day.

Lesson distill (cross-session knowledge distillation)

Periodically, triggered by the user after significant sessions, the system performs a deeper reflection that extracts reusable lessons from accumulated operational experience and proposes updates to skills, knowledge-base entries or decision rules. These proposals are presented to the user for review before being applied; the system never modifies its own skills or knowledge autonomously. Together, session distill and lesson distill form a closed loop: operational

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456 experience generates lessons, lessons improve skills and improved skills produce better
457 operational outcomes.

458 ### *Per-agent memory specialization*

459 Maria and Annika use the same memory infrastructure but store different kinds of information.
460 Maria's memory is primarily scientific: the three-tier knowledge base, evaluated literature,
461 project-level decision context and accumulated domain rules. Annika's memory is primarily
462 operational: run history, workflow state, environment-specific behavior (which software versions
463 are installed, which parameter combinations have been tested, which file formats each tool
464 expects) and tool-specific lessons learned. This separation reflects the agents' different roles and
465 ensures that each agent's persistent memory is optimized for its function.

466 ## Language model backends

467 In the current deployment, both agents use Anthropic Claude Opus 4.6 [37] as the primary model
468 for multi-step tool use and long-context reasoning. Maria can additionally use OpenAI GPT-5.4
469 [38] for complementary scientific consultation. Lightweight operational tasks, including session
470 summarization, context compaction and structured logging, use Qwen 2.5 7B [39] locally.
471 Backend assignment is a configuration parameter rather than an architectural requirement, and
472 model identifiers and settings are recorded in session logs where available.

473 ## Deployment and approval gates

474 The current deployment runs on a single Apple Mac mini (M4, 16 GB unified memory). Agent
475 sessions, tool execution, inter-agent messaging and structural-biology software run locally on
476 macOS with Apple Silicon. Primary LLM inference uses external API providers; the local

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machine does not require a GPU cluster. The user interacts through the OpenClaw messaging layer, including Telegram, WhatsApp or other chat clients.

The deployed structural-biology stack included ChimeraX [40], ISOLDE [7], PHENIX [5], Coot [41], Servalcat [8]/REFMAC5 [9], gemmi and task-specific scripts.

Approval gates are applied to strategy-level changes, ligand-orientation choices, ion reassignments, residue mutations, broad refinement-route changes and any deposition- or release-related action. Annika may run preflight checks, format conversions, validation scripts and retry attempts that do not change scientific model content. Major model-changing actions are presented to the user for review before execution.

Agents generated plans, commands, diagnostic summaries and candidate evidence packages, but did not autonomously decide final biological interpretations or deposition-ready coordinates. The author reviewed manuscript claims, case-study outputs, figure content and model-changing decisions. Tool permissions were limited by the local operating environment and by explicit approval gates; potentially destructive file operations, model-changing edits and release actions required human approval.

Case-study validation protocols

For Example 1, validation used Phenix 2.0 (`phenix.validation_cryoem`, `phenix.molprobity`, `phenix.mtriage` and `phenix.emringer`) at 3.8 Å against the deposited EMD-8332 full map; half-maps were not used. PDB 5T0C was excluded from pipeline stages and used only for post hoc comparison. AlphaFold DB v4 monomers [26] were used for all 32 unique subunits. ProSMART [25] distance restraints were generated from AlphaFold-predicted structures for

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Servalcat refinement, and AlphaFold monomers were used as reference models in PHENIX real-space refinement.

For Example 3, ligand restraints for 5-formimidoyltetrahydrofolate were generated with PHENIX/eLBOW. The Site 2 transferase active-site pose was transferred from PDB 1QD1/FON after structural superposition. Site 1 at the inter-domain interface was evaluated by residue-environment mapping, dual-orientation ChimeraX fitmap scoring and per-group density analysis on unrefined seed poses. Flexible fitting used ISOLDE; after the initial OpenMM template failure, a USER_LIG force-field XML was generated through antechamber, tleap and ParmEd. Final refinement used PHENIX real-space refinement with individual isotropic ADP and custom pterin planarity restraints. Validation included phenix.mtriage, per-group ligand density, B-factor parity and HOLE v2.3.2 channel profiling. The map and model files are withheld because the biological structure is from an ongoing study.

For Example 2, Annika extracted coordination shells for all 530 deposited metal positions in PDB 8B0X with gemmi, computed BVS values for user-specified candidate ions and assigned confidence grades from BVS, coordination number and geometry. Corrected sites were refined with REFMAC5 through Servalcat using bond-distance restraints and angle-derived 1–3 distance restraints. CMM validation was run on the original 530 positions; after correction, the 9 sites reclassified as water were excluded from corrected ion-score totals, leaving 521 ion sites.

Data Availability

The public structural data used in this study are available from public repositories: EMD-8332 (EMDB), PDB 5GJR, PDB 5T0C, PDB 1QD1 and PDB 8B0X (Protein Data Bank). The cryo-EM map, model and accession information for the ongoing-work ligand example are from an

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ongoing study and are withheld from the initial public resource archive; they can be included in a confidential reviewer-only evidence bundle if requested and will be deposited separately upon publication of that work. Derived models, workflow traces and case-study evidence for Examples 1-3 are organized as reviewer-only evidence bundles for validation of the manuscript claims, not as public tutorial datasets in the initial public archive. Because the case studies demonstrate StructAgent workflows rather than report newly determined biological structures, formal PDB deposition of derived outputs was not pursued.

Code Availability

The StructAgent source repository is available at https://github.com/bhgtiger/StructAgent under an Apache-2.0 license. The manuscript corresponds to commit `cdbf720dbb19061aa94c10e7e52e329b54982854`. A DOI-bearing Zenodo archive has been reserved for the submitted release: `10.5281/zenodo.20015169`; reviewer access to the draft archive is available at https://zenodo.org/records/20015169?preview=1&token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6Ijc3ZjkzNWNmLWQzY2EtNDI3Ny1iYjg4LTQ0NDEzZDBkZjg0YyIsImRhdGEiOnt9LCJyYW5kb20iOiI0ODYyNmM2MmU0Zjg4MTQ4N2RlZDAwN2M5OWY4NjY0MyJ9.mj-JUwE-vcdgzofuSQHZdUy_h7KhitwiZxBj0IEBXBLII8qfSXqJhK9Ns2ltbErxwnPAO2a5SJsZJU66du7J7Q.

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Author Contributions

X.G. conceived the project, designed the system architecture, implemented StructAgent, conducted all experiments, analyzed the data and wrote the manuscript. Large language model assistants were used for manuscript drafting and editing under the author's direction and review.

Competing Interests

The author declares no competing interests.

Figure Legends

Fig. 1

Fig. 1 | StructAgent separates structural-biology reasoning from local software execution. The user supplies scientific intent, constraints, approvals and final interpretation. Maria provides literature-grounded reasoning and validation criteria; Annika manages local execution, retries, logs and provenance. The audit framework combines curated literature and skills, established structural-biology software, approval gates, validation output and provenance records. The intended output is a reviewable evidence package for expert decision-making.

Fig. 2

Fig. 2 | Whole-assembly rebuilding of the 26S proteasome from an earlier structural template. (a) StructAgent pipeline for refitting PDB 5GJR into EMD-8332. Recovery events are annotated. **(b)** Validation comparison among the rigid-body 5GJR template fit, the StructAgent

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645 rebuild and deposited 5T0C, all evaluated against EMD-8332 (3.8 Å full map; half-maps not
646 used). Metric direction: higher is better for CC, EMRinger and favored percentages; lower is
647 better for outliers, clashscore, RMSD and FSC resolution; Rama-Z is interpreted relative to zero.
648 **(c)** Per-chain map-correlation improvement for all 64 chains relative to the template fit (mean
649 ⊗CC = +0.24, median = +0.23). **(d)** Local density comparisons for Rpn5 (chain 3, ⊗CC = +0.30)
650 and Rpt2 (chain L, ⊗CC = +0.44). Orange, 5GJR template fit; blue, StructAgent rebuild.

651 **Fig. 3**

652 **Fig. 3 | Systematic metal-ion audit and correction of a 1.55 Å cryo-EM ribosome. (a)** BVS
653 classification of all deposited ion sites in PDB 8B0X before correction (n = 530) and after
654 correction and refinement (n = 521 ion sites; 9 original positions reclassified as water are
655 excluded from corrected ion-score totals). Confidence grades A-F are assigned from BVS,
656 coordination number and geometry. **(b)** Independent rescoring with the CheckMyMetal (CMM)
657 validation server [32] across the original 530 positions; corrected-model ion-score totals exclude
658 the 9 water-reclassified positions. Top-scoring sites increase from 69% to 79%, and score-0
659 outliers decrease from 15 to 4. **(c)** Global refinement and ion-coordination statistics before and
660 after correction plus REFMAC5 refinement with bond-distance and angle-derived 1-3 distance
661 restraints. **(d)** Three representative sites before (deposited 8B0X, top) and after (corrected +
662 refined, bottom): A/1622 and A/1707 (correctly assigned Mg^{2+}, with improved coordination
663 geometry after refinement) and A/3044 ($Mg^{2+} \rightarrow K^{+}$ reclassification, one of four sites previously
664 documented as misassigned by Majorek et al. [29]). BVS, bond-valence sum; CMM,
665 CheckMyMetal; FSC_work, work Fourier shell correlation; gRMSD, geometric root-mean-
666 square deviation; ⊗, corrected minus deposited.

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667 ## Fig. 4

668 **Fig. 4 | Folate-ligand fitting into cryo-EM density with scientific reasoning. (a) Workflow**
669 **for fitting 5-formimidoyltetrahydrofolate into two sites in an ongoing-work enzyme.**
670 **Annika's execution steps are shown with Maria/user consultation points. (b) Site 1, the**
671 **inter-domain site without a homologous ligand pose, after orientation selection, ISOLDE**
672 **fitting and PHENIX refinement; per-ligand CC = 0.68. (c) Site 2, the transferase active site**
673 **fitted by homolog transfer from PDB 1QD1/FON and refined with the two-site model; per-**
674 **ligand CC = 0.67. The glutamate tail shows weak density consistent with conformational**
675 **disorder at approximately 3.2 Å local resolution. The map and model are from an ongoing**
676 **biological study and will be deposited separately.**

677 ## Fig. 5

678 **Fig. 5 | StructAgent agent architecture. (a)** Maria's domain-specialist architecture, including
679 literature discovery, paper digestion, inter-agent consultation, the three-tier knowledge base and
680 scientific memory. **(b)** Annika's execution-specialist architecture, including tool-interface skills,
681 workflow-composition skills, run state, validation metrics and retry logic. Approval gates are
682 applied to strategy changes, model-changing corrections and deposition-related actions.
683 Resource-release boundaries are described in the Data Availability and Code Availability
684 sections.

685 ## Fig. 6

686 **Fig. 6 | Context separation in single-agent and two-agent designs.** In a single-agent design,
687 accumulated execution detail competes with domain-level reasoning in the same context window.
688 StructAgent keeps execution history in Annika's session and sends focused scientific questions

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689 to Maria. This figure illustrates the design rationale; quantitative ablation of context separation

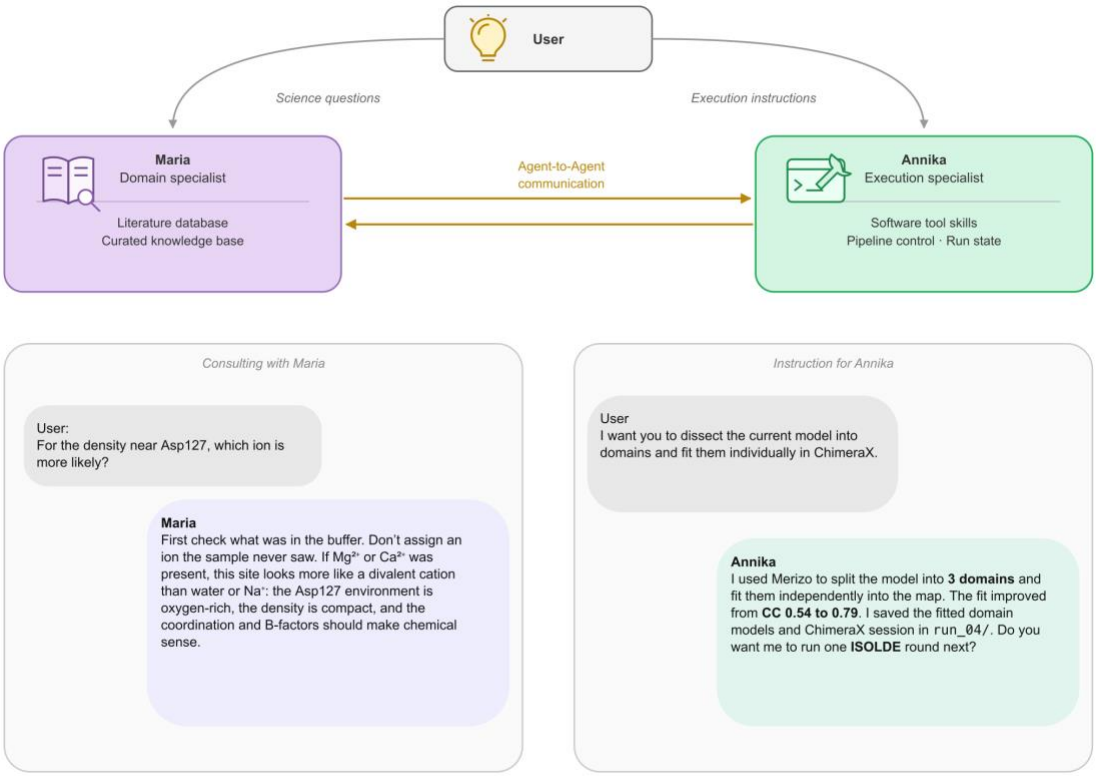
690 was not performed.

691

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Main Figures

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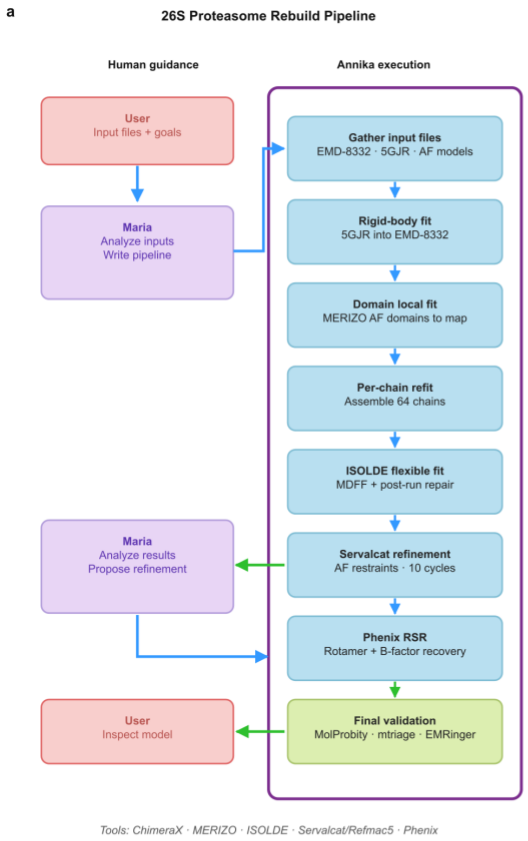


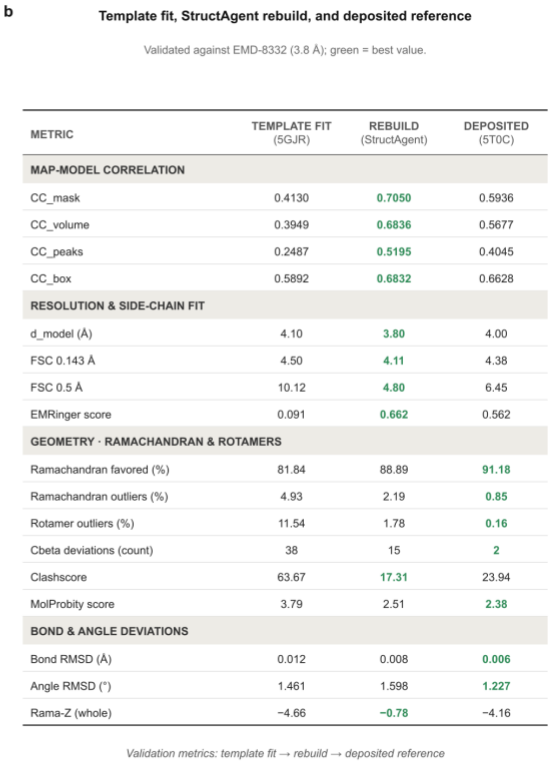
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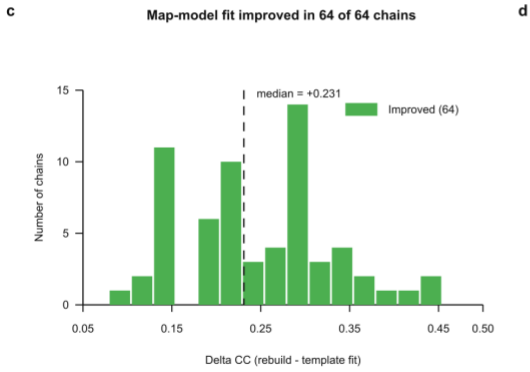
695 *Fig. 1*

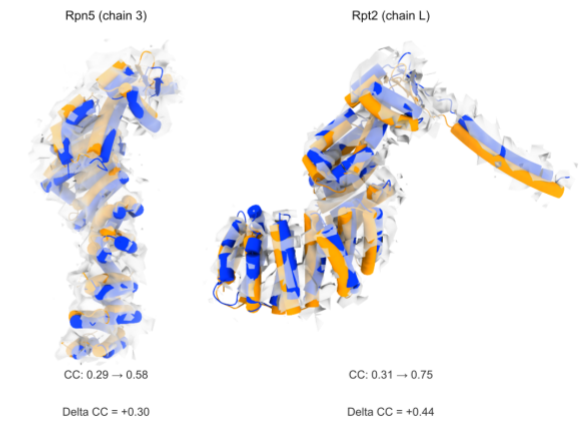
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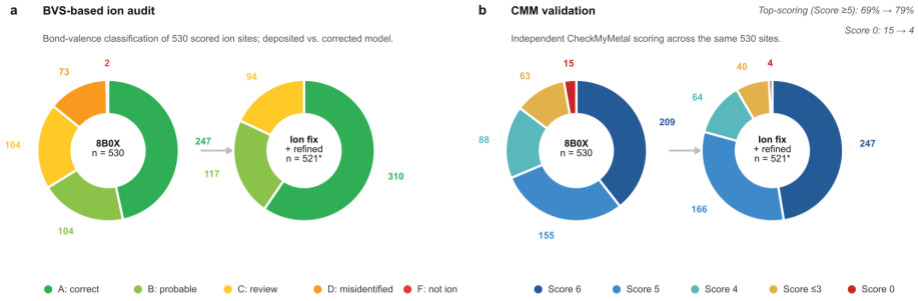


697

698 *Fig. 2*

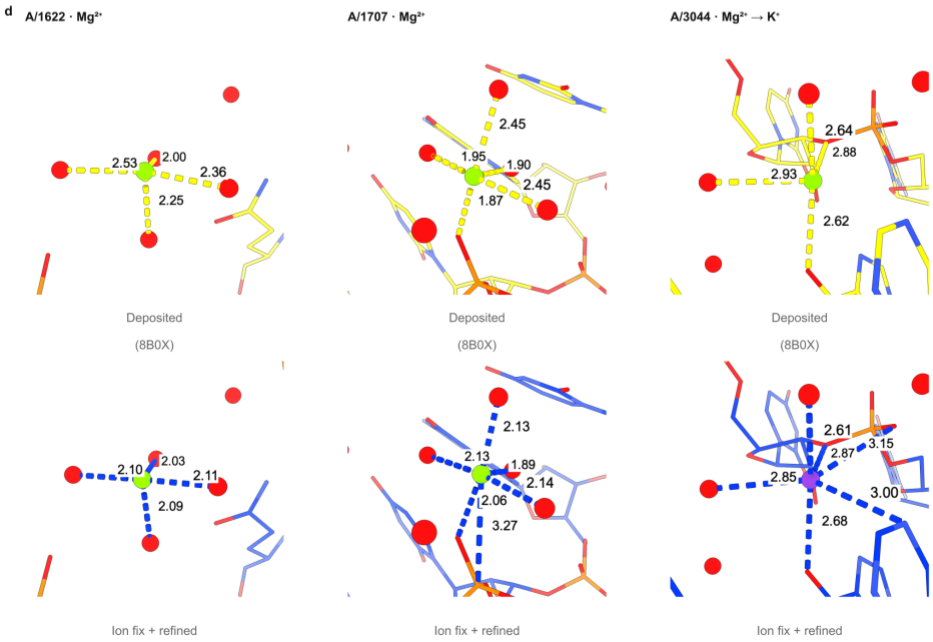
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c **Validation before and after correction + refinement**

METRIC	8B0X	ION FIX + REFINED	Δ
GLOBAL REFINEMENT			
R-factor	0.3836	0.3443	−0.039 −10.2%
FSCwork (map–model)	0.7864	0.8342	+0.048 +6.1%
Bond RMSD (Å)	0.020	0.009	−0.011 −55%
Angle RMSD (°)	1.379	1.172	−0.207 −15%
ION COORDINATION GEOMETRY — SAME-IDENTITY SITES			
Mg^{2+} mean |Δ|→ideal distance (Å)	0.081	0.062	−0.019 −23%
Mg^{2+} distance σ (Å)	0.106	0.088	−0.018 −17%
Mg^{2+} gRMSD (°)	8.12	7.80	−0.32 −3.9%
Zn^{2+} mean |Δ|→ideal distance (Å)	0.166	0.158	−0.008 −5.0%
Zn^{2+} gRMSD (°)	26.67	26.44	−0.23 −0.8%
K^{+} mean |Δ|→ideal distance (Å)	0.232	0.225	−0.007 −3.0%

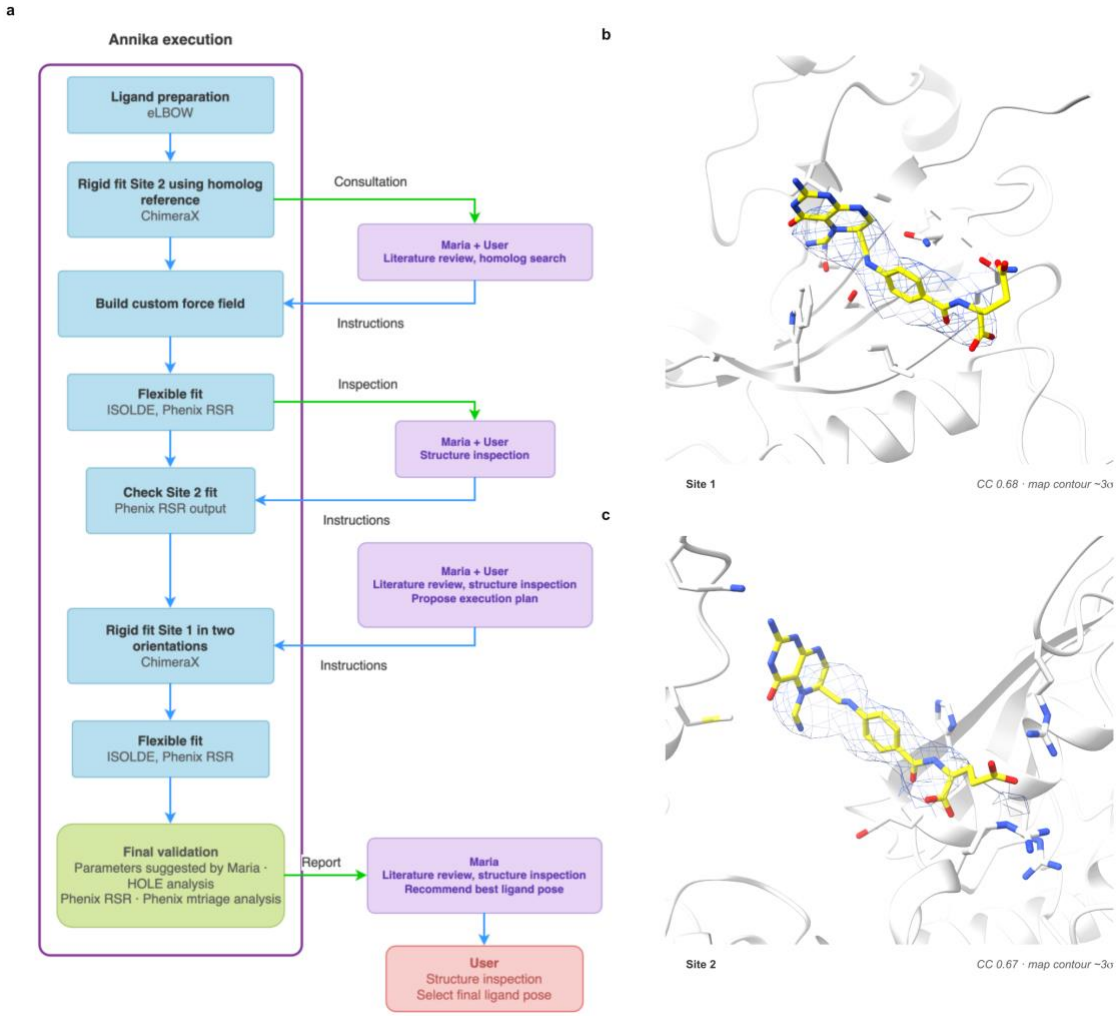


700

701 *Fig. 3*

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704 *Fig. 4*

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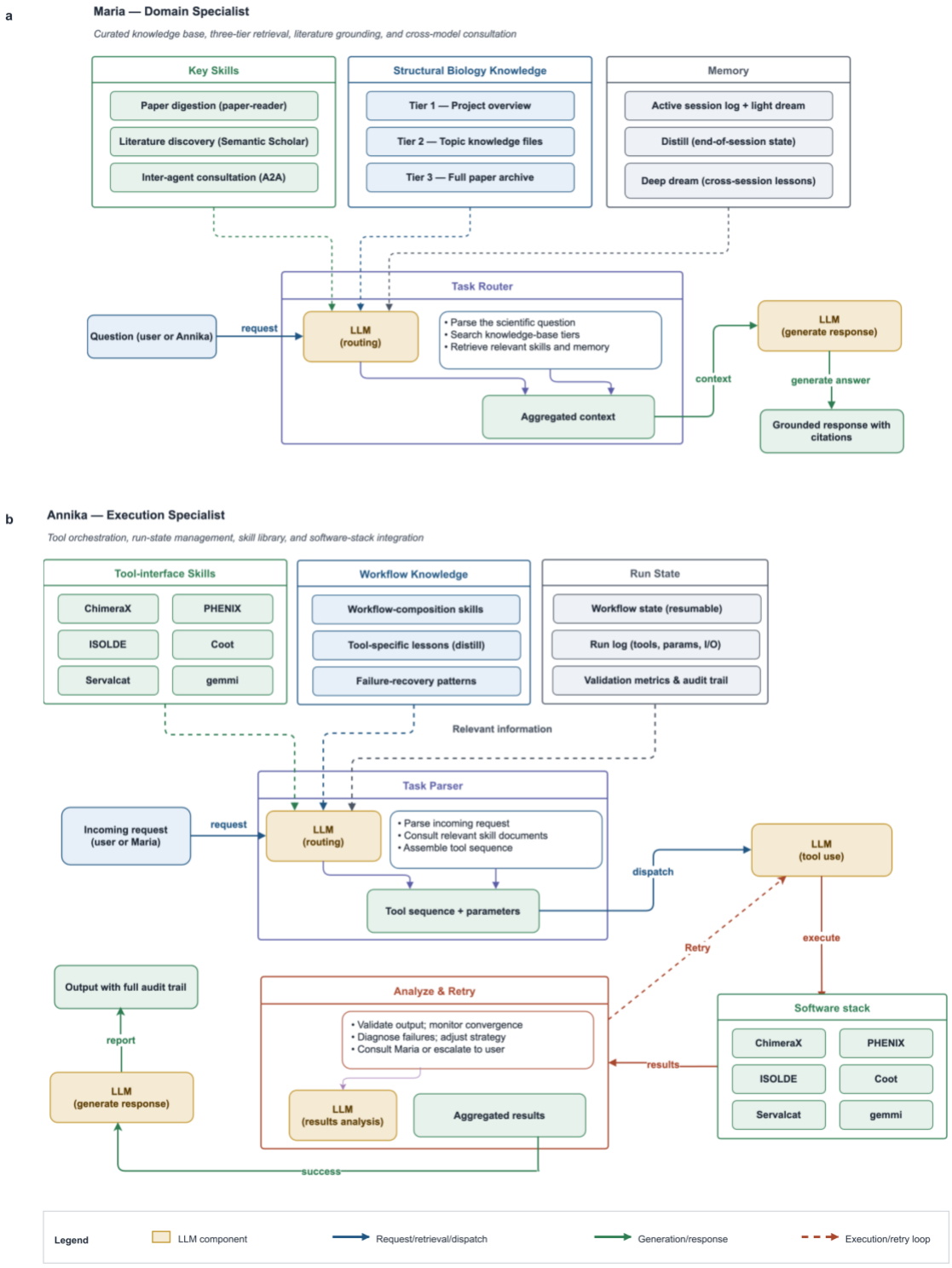


Fig. 5

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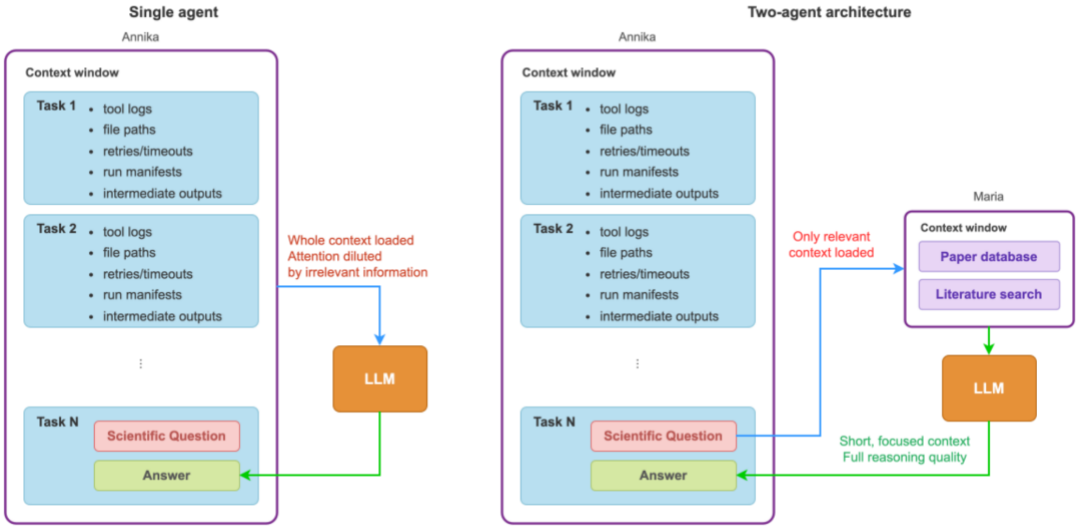


Fig. 6

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Supplementary Tables

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Supplementary Table 1 — StructAgent skill inventory

Skills are structured protocol documents that define how each agent handles a class of tasks. Each skill specifies the task scope, required tools, execution steps, validation criteria, and failure-recovery logic. Skills evolve through the reviewed distillation mechanisms described in Online Methods.

Maria — Domain-specialist skills

Skill	Purpose	Key tools / resources	Typical trigger
Paper reader	Reads primary research papers and produces structured digests with critical assessment	`pdf_split.py` extraction, section detection, metadata parsing, figure extraction	"Read this paper"; primary research PDF provided in a project context
Review paper	Reads review articles, scoping reviews, systematic reviews, and meta-analyses; evaluates synthesis quality, scope, taxonomy, claim–evidence anchors, and follow-up primary literature	Same extraction pipeline as Paper reader, with a review-specific digest template for evidence auditing	"Read this review"; review or meta-analysis PDF provided
Database	Queries, searches, cross-references, and answers questions from project paper databases; identifies literature gaps	Project YAML databases, cross-reference index, connection maps	"What papers do we have on X?"; "Any gaps in the database?"
Discovery	Performs gap-driven literature expansion: identifies candidate papers, deduplicates against the existing database, triages relevance and quality, and produces a ranked reading queue	Semantic Scholar API for search, recommendations, and citation-graph traversal; project database for deduplication	"Find more papers on X"; "Expand the literature around Y"
Distill	Performs end-of-session reflection: updates working memory, proposes reusable lessons for skill/protocol improvement, and writes a session summary	Session interaction trace, working-memory files, skill files	"Distill"; "Ready for reset"; "Summarize session"

Annika — Execution-specialist skills

Skill	Purpose	Key tools / resources	Typical trigger

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Skill	Purpose	Key tools / resources	Typical trigger
ChimeraX	Automates UCSF ChimeraX for fitting, superposition, measurements, model editing, docking preparation, and visualization	ChimeraX CLI (`--nogui --script` for batch work; GUI mode when needed), structured JSON output parsing	"Fit the model into the map"; "Superpose these structures"; "Run ChimeraX"
ISOLDE	Runs interactive flexible fitting (MDFF) of models into cryo-EM or crystallographic maps; manages simulations, restraints, ligand handling, validation, and export to PHENIX-compatible outputs	ChimeraX with the ISOLDE plugin (GUI mode required), REST/monitoring bridge, convergence-tracking scripts	"Run ISOLDE"; "Flexible fitting"; "Fix this region interactively"
PHENIX	Executes refinement and validation workflows for X-ray crystallography and cryo-EM; supports ligand restraint generation, model preparation, metal coordination restraints, and geometry validation	PHENIX CLI (`phenix.refine`, `phenix.real_space_refine`, `phenix.ready_set`, `phenix.reduce`, eLBOW, MolProbity)	"Refine the model"; "Run real-space refinement"; "Validate geometry"
Coot	Performs local model rebuilding, ligand/monomer handling, density-guided cleanup, water finding/pruning, dictionary/restraint management, and Coot-specific scripting	Coot 1 CLI (`coot --no-graphics --script` for automation; GUI mode for interactive tasks), Python scripting API	"Rebuild this region in Coot"; "Add waters"; "Fit the ligand"
CCP4	Executes explicitly requested CCP4 programs, including reciprocal-space refinement, cryo-EM refinement, ligand restraint generation, MTZ inspection/preflight, and FreeR handling	CCP4 suite: REFMAC5/`refmacat`, Servalcat, AceDRG, `freerflag`, `mtzdump`, `cad`, `sftools`, `pdbset`, `pdbcur`, Phaser, MOLREP, Buccaneer, Nautilus, AIMLESS, POINTLESS, CTRUNCATE, Privateer; explicit-trigger only, with no automatic file selection	"Run REFMAC5"; "Servalcat refine_spa"; "AceDRG for this ligand"; named CCP4 tool or `.com` script
Rosetta EMERALD	Executes density-guided ligand docking into cryo-EM maps using Rosetta GALigandDock with `beta_genpot` density-weighted scoring; used when no trustworthy transferred ligand pose exists	Rosetta ≥2023.06 (`rosetta_scripts.*`), GALigandDock XML protocol, ligand `.params` generation via AmberTools/Antechamber or `mol2genparams.py` with AM1-BCC charges	"Run EMERALD"; "Rosetta EMERALD"; "Dock ligand into a cryo-EM map with Rosetta"; "GALigandDock with density"
Structural build	Orchestrates multi-step structure-building pipelines and routes subtasks to the appropriate execution skills	ChimeraX, ISOLDE, PHENIX, Coot, CCP4, Rosetta EMERALD, Merizo for domain segmentation, and supporting scripts	"Build the structure"; "Fit and refine end-to-end"
Structural strategy	Provides decision-level planning for structure building: what to do, in what order, and why; covers fitting, model building, refinement, validation, and special cases	Domain decision trees, workflow heuristics, validation gates, and consultation with Maria when scientific judgment is required	"What should I do next?"; "How should I approach this map?"
Distill / Dream	Runs the two-lane reflection process: lightweight session-state preservation (`dream_light`, corresponding to session distill) and deeper skill-improvement proposals (`dream_deep`, corresponding to lesson distill)	Session interaction trace, skill files, learned-lesson files, working-memory state	"Distill"; "Dream light"; "Dream deep"; "Session distill"; "Lesson distill"; end-of-session review

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Notes

1. Each skill document is human-readable and can be inspected by the user at any time.
2. Skills accumulate lesson and learned files that record environment-specific knowledge discovered during use, such as ChimeraX command quirks, PHENIX parameter defaults, ISOLDE input requirements, CCP4 setup behavior, and Rosetta build/version constraints.
3. The skill inventory is extensible. New skills can be added by writing a new protocol document and, when needed, supporting scripts. Existing skills evolve through reviewed distillation.
4. CCP4 and Rosetta EMERALD are execution-only skills. Strategy decisions remain in **Structural strategy**, which selects the appropriate computational route and then dispatches the chosen command to the matching execution skill. This separation keeps the decision layer engine-agnostic.
5. Maria and Annika can run on different LLM backends. In the current deployment, both agents use Anthropic Claude Opus 4.6 as the primary model; Maria can additionally use OpenAI GPT-5.4 for complementary scientific consultation, and Qwen 2.5 7B is used for lightweight operational tasks such as summarization, logging, and context compaction. Backend assignment is a configuration choice rather than a system requirement; see Online Methods for details.
6. Online Methods uses the descriptive terms **session distill** and **lesson distill**. In Annika's deployed skill library, these correspond to the `dream_light` and `dream_deep` lanes of the **Distill / Dream** skill.

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Supplementary Table 2 — Knowledge base paper inventory

This table lists all papers currently in StructAgent's curated knowledge base, organized by category. Papers are stored as structured YAML entries with metadata, structured digests with critical assessments, and cross-reference connections. The knowledge base feeds Maria's three-tier retrieval system (see Methods): Tier 1 (project briefs) and Tier 2 (synthesized topic files) are distilled from these sources; Tier 3 provides searchable access to individual digests and metadata.

Interactive model building and fitting

First author	Year	Title	Journal	Key contribution to knowledge base
Croll	2018	ISOLDE: a physically realistic environment for model building into low-resolution electron-density maps	Acta Cryst. D	Core ISOLDE skill: MD-based interactive fitting, validation integration, failure modes at low resolution
Croll	2021	Adaptive conformational restraints for interactive model rebuilding	Acta Cryst. D	Tuneable top-out/von Mises restraint design; directly informs ISOLDE restraint parameters
Emsley	2004	Coot: model-building tools for molecular graphics	Acta Cryst. D	Original Coot architecture and algorithms; historical baseline
Emsley	2010	Features and development of Coot	Acta Cryst. D	Coot feature catalog: RSR, rotamer fitting, water finding, ligand fitting, validation suite
Casañal	2020	Current developments in Coot for cryo-EM and crystallographic data	Protein Science	Coot cryo-EM updates: jiggle fit, Geman-McClure restraints, register-shift nudging
Emsley	2021	Interactive model building and refinement with Coot 1.0 (workshop talk)	CCP-EM workshop	Practical Coot 1.0 insight: interactive refinement weight control, shift-field B-factors, difference maps

Automated model building (de novo)

First author	Year	Title	Journal	Key contribution to knowledge base
Jamali	2024	Automated model building and protein identification in cryo-EM maps	Nature	ModelAngelo: GNN-based building + protein identification via HMM. Performance benchmark baseline

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First author	Year	Title	Journal	Key contribution to knowledge base
Su	2026	CryoAtom improves model building for cryo-EM	Nat. Struct. Mol. Biol.	AF2-inspired local attention transformer; improved completeness over ModelAngelo
Zhang	2026	Accurate macromolecular complex modeling with CryoZeta	bioRxiv	Diffusion-based building on AF3 architecture; protein + nucleic acid, low resolution
Terashi	2023	DeepMainmast: integrated protocol for cryo-EM with deep learning and structure prediction	Nature Methods	DL atom detection + AF2 integration; constraint programming for chain ID assignment
Terashi	2018	De novo main-chain modeling for EM maps using MAINMAST	Nat. Commun.	Graph-based (MST + tabu search) de novo building; historical pre-DL baseline
Pfab	2021	DeepTracer for fast de novo cryo-EM protein structure modeling	PNAS	U-Net backbone prediction + TSP tracing; 2nd-generation DL builder
Cowtan	2006	Buccaneer: automated model building—tracing protein chains	Acta Cryst. D	Foundational Cα-likelihood chain tracing; still used in CCP4/CCP-EM pipelines
Terwilliger	2018	A fully automatic method yielding initial models from high-resolution cryo-EM maps	Nature Methods	Phenix map_to_model: zero-parameter auto-build pipeline; benchmark baseline for all DL methods
Chojnowski	2021	Accuracy of protein models automatically built into cryo-EM maps with ARP/wARP	Acta Cryst. D	Systematic 119-map benchmark; local resolution determines buildability, not global FSC
Chojnowski	2022	findMySequence: neural-network-based identification of unknown proteins	IUCrJ	Sequence identification from main-chain-only models + maps; found register errors in deposited structures
Wang	2023	CryoREAD: de novo structure modeling for nucleic acids in cryo-EM	Nature Methods	First DL method for automated DNA/RNA building; cascaded U-Net + VRP backbone tracing
Hryc	2022	Pathwalking: next-generation de novo model building utilities	Biomolecules	TSP-based building + semi-automated ligand/water identification
Bansia	2025	Deep learning-based automated model building methods in cryo-EM (review)	Biochem. Soc. Trans.	Comprehensive review of ~50 DL methods (2017–2025); landscape and gap analysis

Refinement

First author	Year	Title	Journal	Key contribution to knowledge base
Afonine	2018	Real-space refinement in PHENIX for cryo-EM and crystallography	Acta Cryst. D	Phenix RSR engine: atom-centered target, auto weight, SA; core of Phenix skill
Murshudov	2011	REFMAC5 for the refinement of macromolecular crystal structures	Acta Cryst. D	REFMAC5 reference: ML likelihood, jelly-body, TLS, KL-divergence ADP restraints

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First author	Year	Title	Journal	Key contribution to knowledge base
Murshudov	2016	Refinement of atomic structures against cryo-EM maps	Methods Enzymol.	Bayesian cryo-EM refinement theory; why crystallographic shortcuts are dangerous in EM
Nicholls	2018	Current approaches for fitting and refinement using CCP-EM	Acta Cryst. D	CCP-EM/REFMAC5 cryo-EM pipeline: reciprocal-space ML, jelly-body, Divide and Conquer
Yamashita	2021	Servalcat: cryo-EM single-particle structure refinement and map calculation	Acta Cryst. D	REFMAC5 wrapper for cryo-EM: half maps → ML refinement → Bayesian Fo-Fc difference maps
Yamashita	2023	GEMMI and Servalcat restrain REFMAC5	Acta Cryst. D	Restraint-generation refactor; alias mechanism fixing modified residue nomenclature
Wang	2016	Automated structure refinement of macromolecular assemblies using Rosetta	eLife	Fragment-based MC refinement; catches backbone errors gradient methods miss; 20,000× slower
Zubatyuk	2025	AQuaRef: machine learning accelerated quantum refinement	Nat. Commun.	ML potential replacing library restraints; paradigm shift in refinement. Protein-only for now
Kim	2019	cryo_fit: democratization of flexible fitting for cryo-EM	J. Struct. Biol.	Phenix-integrated MD flexible fitting via GROMACS; automation of MDFF setup
Kidmose	2019	Namdinator: automatic molecular dynamics flexible fitting	IUCrJ	MDFF + Phenix RSR in one-click pipeline; convenience wrapper
Wang	2012	Real-space refinement with DireX	Biopolymers	DEN restraint-based fitting; pre-revolution baseline, now superseded

Validation

First author	Year	Title	Journal	Key contribution to knowledge base
Chen	2010	MolProbity: all-atom structure validation	Acta Cryst. D	Foundational all-atom validation: REDUCE, PROBE, clashscore, Ramachandran/rotamer analysis
Prisant	2020	New tools in MolProbity: CaBLAM for cryo-EM backbone	Protein Science	CaBLAM for low-resolution cryo-EM; checks multi-residue backbone logic
Richardson	2023	Pernicious forms of overfitting versus good new tools	Acta Cryst. D	Strongest critique of sparse-data overfitting; CaBLAM corrective logic
Barad	2015	EMRinger: side-chain-directed model and map validation	Nature Methods	Chi1-vs-density validation bridging geometry and map-model fit
Pintilie	2025	Q-score as a reliability measure for atomic coordinate models from 3DEM	Acta Cryst. D	Q-score at archive scale; resolution-dependent percentile expectations
Afonine	2018	New tools for the analysis and validation of cryo-EM maps and atomic models	Acta Cryst. D	Phenix validation: d99, dmodel, CC triplet (CCmask/CCvolume/CCpeaks);

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First author	Year	Title	Journal	Key contribution to knowledge base
				archive survey
Ramírez-Aportela	2021	FSC-Q: CryoEM map-to-atomic model quality validation	Nat. Commun.	Local FSC-based map-model fit; detects overfitting into noise/detergent invisible to Q-score
Wang	2022	Validation analysis of EMDB entries	Acta Cryst. D	EMDB three-tier validation framework; archive-scale operational system
Kleywegt	2024	Community recommendations on cryo-EM data archiving and validation	IUCrJ	Community white paper: mandatory half-maps, mask deposition, reporting standards
Lander	2024	Single-particle cryo-EM map and model validation	Curr. Opin. Struct. Biol.	2024 validation landscape: preferred orientation, MolProbity/map-model disconnect, AI density modification risks
Rosenthal	2016	Testing the validity of single-particle maps	Methods Enzymol.	Map validation foundations: tilt-pair, noise substitution, FSC, local resolution
Lawson	2021	Cryo-EM model validation recommendations (2019 EMDataResource challenge)	Nature Methods	Benchmark: metric clustering shows correlation metrics are counterintuitive; 4 recommendations
Lawson	2024	Outcomes of the EMDataResource cryo-EM ligand modeling challenge	Nature Methods	Ligand validation benchmark: composite LIVQ5 + MolProbity best; CaBLAM most discriminating
Gore	2017	Validation of structures in the Protein Data Bank	Structure	Canonical PDB validation reference; all metrics cataloged; X-ray quality trends
Warshaman age	2022	EMDA: a Python package for electron microscopy data analysis	J. Struct. Biol.	Local correlation framework, likelihood-based map overlay, magnification refinement
Pražnikar	2019	Validation using complex network analysis	Sci. Rep.	Graph-theory validation (node degree, graph energy); orthogonal proof-of-concept

Map processing and reconstruction

First author	Year	Title	Journal	Key contribution to knowledge base
Jakobi	2017	Model-based local density sharpening of cryo-EM maps (LocScale)	eLife	Local amplitude scaling without phase transfer; B-factors encode local resolution
Terwilliger	2020	Density modification of cryo-EM maps	Acta Cryst. D	Density modification using prior knowledge; ensemble models to control model bias
Sanchez-Garcia	2021	DeepEMhancer: deep learning for cryo-EM volume post-processing	Commun. Biol.	3D U-Net sharpening + masking without atomic model; enabled de novo loop tracing
Levy	2025	CryoDRGN-AI: neural ab initio	Nature Methods	Fully ab initio heterogeneous

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First author	Year	Title	Journal	Key contribution to knowledge base
		reconstruction		reconstruction; upstream of model building
Zhong	2021	CryoDRGN: reconstruction of heterogeneous structures using neural networks	Nature Methods	VAE-based continuous heterogeneity reconstruction; foundational method

Software suites and infrastructure

First author	Year	Title	Journal	Key contribution to knowledge base
Adams	2010	PHENIX: a comprehensive Python-based system for macromolecular structure solution	Acta Cryst. D	Canonical Phenix reference (v1.5); full pipeline architecture
Liebschner	2019	Macromolecular structure determination: recent developments in Phenix	Acta Cryst. D	2019 Phenix reference: cryo-EM tools, CaBLAM, ISOLDE/Amber/Rosetta integration
Agirre	2023	The CCP4 suite: integrative software for macromolecular crystallography	Acta Cryst. D	2023 CCP4 reference: modernized ecosystem with AlphaFold support, Modelcraft, CCP4 Cloud
Winn	2011	Overview of the CCP4 suite and current developments	Acta Cryst. D	2011 CCP4 reference; historical overview, now superseded by Agirre 2023
Wood	2015	Collaborative computational project for electron cryo-microscopy	Acta Cryst. D	CCP-EM founding paper; community infrastructure for cryo-EM tool integration
Burnley	2017	Recent developments in the CCP-EM software suite	Acta Cryst. D	CCP-EM framework integrating RELION, Flex-EM, TEMPy with unified interfaces
Pettersen	2021	UCSF ChimeraX: structure visualization for researchers, educators, and developers	Protein Science	Comprehensive ChimeraX 1.0 reference; Toolshed/bundle architecture enabling ISOLDE
Goddard	2018	UCSF ChimeraX: meeting modern challenges in visualization and analysis	Protein Science	Early ChimeraX paper; rendering and Toolshed architecture
Young	2017	OneDep: unified wwPDB system for deposition, biocuration, and validation	Structure	PDB deposition infrastructure; 25% coordinate replacement rate

Structure prediction

First author	Year	Title	Journal	Key contribution to knowledge base
Jumper	2021	Highly accurate protein structure prediction with AlphaFold	Nature	AlphaFold2: Evoformer + IPA; paradigm shift for starting models in cryo-EM
Terwilliger	2023	AlphaFold predictions are valuable hypotheses	bioRxiv	Systematic AF vs. experimental comparison: median Cα 0.6 Å but 10% wrong by >2 Å
Jumper	2022	Protein structure predictions to atomic	Nature Methods	Commentary on AF implications;

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First author	Year	Title	Journal	Key contribution to knowledge base
		accuracy with AlphaFold		"amplifier of experimental data" framing

Restraint generation and specialized modeling

First author	Year	Title	Journal	Key contribution to knowledge base
Moriarty	2009	eLBOW: ligand coordinate and restraint generation	Acta Cryst. D	Ligand restraint generator via AM1 QM; the restraint layer feeding Phenix
Long	2017	AceDRG: a stereochemical description generator for ligands	Acta Cryst. D	COD-derived atom typing; hierarchical bond/angle lookup; CCP4 alternative to eLBOW
Babai	2024	Improving refinement with metal-coordination restraints	Acta Cryst. D	95 metal geometry classes via Procrustes; context-aware MetalCoord restraints
de Vries	2021	New restraints and validation for nucleic acid structures in PDB-REDO	Acta Cryst. D	Nucleic acid restraints/validation; ZbpG metric; honest rejection of torsion restraints
Dialpuri	2023	Overall N-glycan conformation analysis and validation in Privateer	Acta Cryst. D	Glycan torsional validation; Z-scores for 13 linkage types; outlier ≠ error
Joosten	2014	PDB_REDO server for macromolecular structure model optimization	IUCrJ	Automated constructive validation pipeline; re-refinement + targeted correction
Lau	2023	Merizo: rapid and accurate protein domain segmentation	Nat. Commun.	DL domain segmentation for Divide and Conquer refinement

Ligand fitting and identification

First author	Year	Title	Journal	Key contribution to knowledge base
Muenks	2023	EMERALD: automatic ligand structure determination guided by cryo-EM	Nat. Commun.	Density-guided genetic algorithm + RosettaGenFF; 57% exact match on 1053 EMDB ligands
Sweeney	2024	ChemEM: flexible docking of small molecules in cryo-EM	J. Med. Chem.	MI density fitting + SA refinement; drug-discovery-facing
Karolczak	2025	Ligand identification in cryo-EM and X-ray maps using deep learning	Bioinformatics	First DL approach to ligand ID in both modalities; reveals cryo-EM standardization gap
Afonine	2021	Automated ordered solvent building (phenix.douse)	Phenix documentation	Water building using geometric + chemical + map-based criteria

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Nucleic acid methods

First author	Year	Title	Journal	Key contribution to knowledge base
Das	2023	Assessment of 3D RNA structure prediction in CASP15	Proteins	First RNA in CASP: DL methods failed; lDDT as universal protein + RNA metric
Kappel	2020	Ribosolve: accelerated cryo-EM determination of RNA-only structures	Nature Methods	DRRAFTER/Rosetta pipeline for RNA model building into cryo-EM maps

Reviews, tutorials, and workshops

First author	Year	Title	Journal	Key contribution to knowledge base
DiMaio	2016	Tools for model building and optimization into cryo-EM density maps	Methods Enzymol.	2016 workflow survey; historical baseline for tool landscape
Clarke	n.d.	Model building and validation for cryo-EM (lecture)	Online lecture	Practitioner's masterclass: failure modes, map carving dangers, craftsman's perspective
Terwilliger	n.d.	Automated model building and map boxing in Phenix (lecture)	Phenix tutorial	Phenix practical workflow: why map boxing, chain-tracing algorithm internals
Vollmar	2024	Current trends in macromolecular model refinement and validation	Acta Cryst. D	CCP4 Study Weekend 2022 editorial; landscape overview
Bock	2024	Single-particle cryo-EM and molecular dynamics simulations	Curr. Opin. Struct. Biol.	MD + cryo-EM synergy review: plunge-freezing, ensemble methods, thermodynamics

Other

First author	Year	Title	Journal	Key contribution to knowledge base
Ippolito	1990	Hydrogen bond stereochemistry in protein structure and function	J. Mol. Biol.	H-bond geometry foundations: syn/anti carboxylate preferences
Fromm	2016	Cryo-EM structure determination using segmented helical reconstruction	Methods Enzymol.	Helical reconstruction; specialized reference
Giri	2024	Cryo2StructData: a large labeled cryo-EM density map dataset	Sci. Data	7,600 labeled cryo-EM maps for AI training; infrastructure dataset

Total: 84 papers spanning model building, refinement, validation, map processing, software infrastructure, structure prediction, restraint generation, ligand handling, nucleic acid methods, and domain reviews.

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765 The knowledge base is not static. Papers are added through Maria's literature discovery skill
766 (Semantic Scholar API-based search, citation-graph traversal, recommendation expansion) and
767 through user-directed reading. Each paper undergoes structured digestion: metadata extraction,
768 critical assessment, method coverage tagging, and cross-reference mapping against the existing
769 database. Synthesized operational rules are distilled into Tier 2 topic files (see Methods)
770 covering validation strategy, refinement decision trees, resolution-dependent parameter
771 guidelines, and tool-specific gotchas.

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Supplementary Table 3 | Tool-level failure and recovery events across the three case studies

ID	Example	Failure trigger / symptom	Detecting agent	Recovery action	User consulted	Outcome
E1-1	Example 1 (proteasome rebuild)	ISOLDE MDFF returned zero forces after hydrogenation	Annika	Diagnosed 237 Å coordinate-frame shift and applied inverse translation	No	Simulation restored; workflow continued
E1-2	Example 1 (proteasome rebuild)	Force-field parameterization gap during rebuild workflow	Annika	Adjusted parameterization path and reran affected step	No	Step recovered
E1-3	Example 1 (proteasome rebuild)	Post-simulation chain drift in zero-density region	Annika	Constrained/realigned affected chain and reran refinement stage	No	Geometry stabilized
E1-4	Example 1 (proteasome rebuild)	File-system case-sensitivity collision	Annika	Renamed/canonicalized file paths and reran step	No	Pipeline resumed
E1-5	Example 1 (proteasome rebuild)	B-factor inflation crashed correlation scoring	Annika	Corrected B-factor handling and reran validation	No	Correlation scoring restored
E2-1	Example 2 (ion audit)	Phenix RSR failed: unrecognized parameter `optimize_weight=True` in `phenix.real_space_refine`	Annika	Removed unsupported parameter and retried with corrected parameter set	No	Subsequent failure exposed next blocker
E2-2	Example 2 (ion audit)	Phenix RSR reference-model approach killed by SIGKILL (out-of-memory) on 16 GB system with 151,507-atom 70S ribosome	Annika	Reduced run scope and adjusted strategy; then pivoted	No	Subsequent failure exposed next blocker
E2-3	Example 2 (ion audit)	Phenix RSR `reference_coordinate_restraints` failed: 283 unknown nonbonded energy types from modified nucleotides without CIF restraints	Annika	Pivoted from Phenix RSR to Refmac5 via Servalcat	Yes (Maria consulted on strategy pivot)	Workflow continued in alternate toolchain
E2-4	Example 2 (ion audit)	Refmac5 EXTE ANGL cascade, step 1: long single-line angle restraint exceeded Refmac5 keyword-line parser length limit	Annika	Reformatted restraints across continuation lines	Yes (Maria consulted during cascade)	Proceeded to next step
E2-5	Example 2 (ion audit)	Refmac5 EXTE ANGL cascade, step 2: continuation-line format failed atom matching ("Number of defined atoms = 2")	Annika	Re-emitted restraints using original PDB chain names	Yes	Proceeded to next step
E2-6	Example 2 (ion audit)	Refmac5 EXTE ANGL cascade, step 3: original chain names did not match Servalcat-prepared CRD internal naming	Annika	Re-emitted restraints using Servalcat-renamed chain identifiers	Yes	Proceeded to next step
E2-7	Example 2	Refmac5 EXTE ANGL	Annika	Switched to `@keyword_file`	Yes	Proceeded

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ID	Example	Failure trigger / symptom	Detecting agent	Recovery action	User consulted	Outcome
	(ion audit)	cascade, step 4: Servalcat-renamed chains (e.g. `3_p`, `3_1` with underscore suffixes) not recognized by Refmac5 EXTE ANGL parser		external-restraint inclusion		to next step
E2-8	Example 2 (ion audit)	Refmac5 EXTE ANGL cascade, step 5: `@keyword_file` route hit the same parser issue — Refmac5 EXTE ANGL cannot resolve underscore-containing nonpolymer chain names; angle restraints not feasible	Annika + Maria	Reformulated angle restraints as 1-3 distance restraints via law-of-cosines equivalence; generated 3,676 EXTE DIST entries (`generate_13dist_restraints.py`, `angles_to_13dist.py`)	Yes	Refmac5 Run 2 completed; refinement proceeded
E3-1	Example 3 (ligand fitting)	OpenMM lacked template for formimidoyl-THF dianion; ISOLDE run failed silently	Annika	Built custom parameters via antechamber + tleap + ParmEd and retried	No	ISOLDE converged; ligand fitting completed